

Root Cause Analysis (RCA): Uber's Decline in DAUs

Problem Statement

"There has been a **10% decline in Uber's Daily Active Users (DAUs)** over the last 7 days across major markets globally. This decline has no specific demographic or regional segmentation and was sudden."

Clarifying Questions

System Factors (Sanity Checks)

- **Q: Has the definition of DAU changed?**
  - No.
- **Q: Are metrics being calculated correctly?**
  - Yes, verified by the analytics team.
- **Q: Has Uber experienced any technical issues like system downtime?**
  - No significant system downtime was reported.

External Factors

- **Q: Is the decline unique to Uber or industry-wide?**
  - The decline appears specific to Uber. Competitors like Lyft and Ola reported steady DAUs.
- **Q: Any competitor campaigns or pricing changes?**

    - Lyft recently launched an aggressive promotional discount campaign targeting key Uber markets.
- **Q: Any macroeconomic factors influencing user behavior?**
    - No major changes observed; consumer spending trends remain stable.

Out-of-Control Factors

- **Q: Were there any natural disasters or global events impacting users?**
    - None reported.
- **Q: Any regulatory or political changes affecting Uber's operations?**
    - No new regulations impacting ride-hailing services.

Internal Factors

- **Q: Did Uber launch new features or make updates during this period?**
    - Yes, a major update to the driver and rider app was released last week.
- **Q: What changes were made in the update?**
    - Improved app design for both drivers and riders, changes to ride-matching algorithms, and new verification protocols for safety.
- **Q: Did the update disrupt any core functionalities?**
    - Yes, some users reported ride cancellations due to extended matching times and delayed driver assignment.
- **Q: Are there customer complaints about these issues?**
    - Yes, a significant uptick in complaints about ride cancellations, longer wait times, and app crashes was noted.
- **Q: Were these issues isolated to specific platforms (e.g., iOS, Android)?**
    - Complaints were higher among Android users, accounting for 70% of total complaints.
- **Q: Did the app update introduce new login or authentication steps?**
    - Yes, a new two-factor authentication was implemented, requiring an OTP for login, which delayed ride bookings.
- **Q: Any reported delays in OTP delivery?**
    - Yes, delays in OTP delivery were observed, particularly during peak hours.

Root Cause Identified

The primary issue stems from **technical friction introduced by the app update**, specifically related to:

1. Delayed driver assignment due to algorithm changes.
2. Increased ride cancellations caused by extended wait times.
3. Disruption of the login process due to OTP delays.

Potential Impact

- Affected ~10% of DAUs, including both riders and drivers.
- Resulted in increased user dissatisfaction, leading to churn.

Solutions

Immediate Actions:

1. Address complaints by providing faster support to riders and drivers facing OTP or ride cancellation issues.
2. Rollback or optimize the ride-matching algorithm to improve driver assignment speed.
3. Send push notifications to riders with pending OTP verifications, providing incentives to complete the process.

Long-Term Mitigation Strategies:

1. Scale server capacity to handle OTP requests during peak times.
2. Conduct A/B testing for future algorithm changes to minimize disruptions before a full rollout.
3. Improve communication with users about app updates and provide a grace period for mandatory authentication changes.

Monitoring and Continuous Improvement:

- Track key metrics like ride completion rates, wait times, and user complaints over the next quarter.
- Monitor churn rates and DAU recovery post-implementation of fixes.
- Conduct regular user feedback sessions to understand pain points caused by new updates.

Summary

The recent **app update**, aimed at improving user experience and safety, inadvertently caused disruptions in ride matching and login processes, leading to a **10% decline in DAUs**. By addressing technical issues and improving the rollout process for future updates, Uber can restore user trust and engagement while preventing similar issues.